Beyond The @Mention

When sentiments tell a product story

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Presentation Outline

01 **Executive Summary**

02 **Business and Data Understanding**

03 **EDA**

04 **Modelling**

05 **Insights**

06 **Recommendations**

Project Overview

Business Case.

- Social media provides real-time customer feedback, making sentiment analysis valuable for product development, customer support, and brand monitoring.

Business Case

- Transform massive volumes of unstructured, noisy social media text into structured, actionable business intelligence.

Key Objectives.

- Baseline Development.
- Benchmark Performance.
- Interative Optimization
- Propose Retention strategy

> *Turning every emotion into business insights .*

Load, Explore, Clean

Load.

- Necessary libraries were loaded .
- The dataset “Tweet_NLP” was used for the analysis
- Python Language on Jupyter Notebook platform.

Explore

- The data had 9,093 rows and 3 columns.
- Data had float, float and integers .
- Duplicates(22) and missing data (5,082)

Clean.

- Dropping redundant columns
- Converting into customer usage patterns and rations
- Adding missing columns.
- Creating column to check customer behaviour

> *Analyze what you know!*

Data Exploration

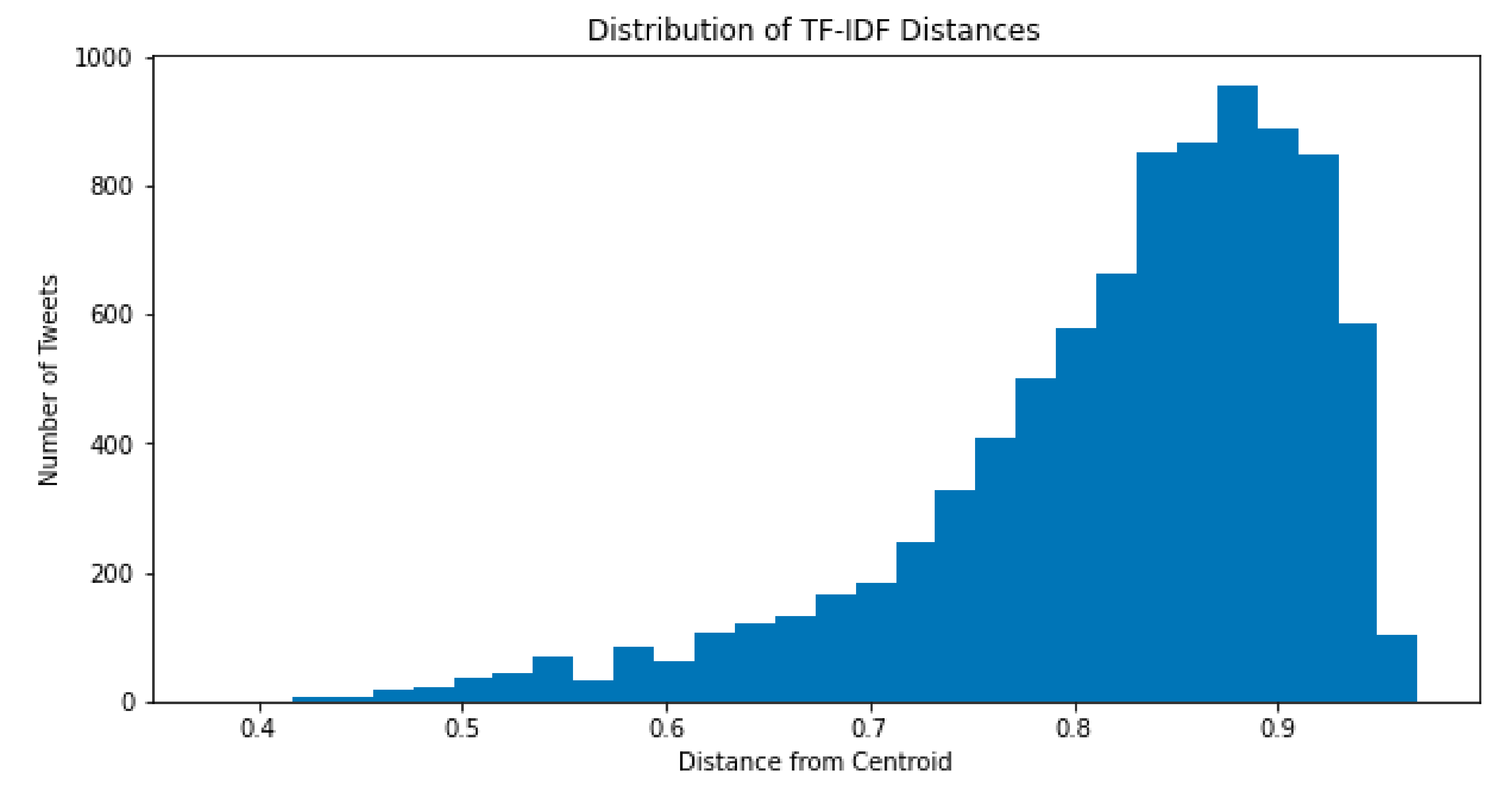
Word Cloud

High Engagement Around Events and Locations (SXSW & Austin).

Apple Ecosystem Dominates the Conversation.

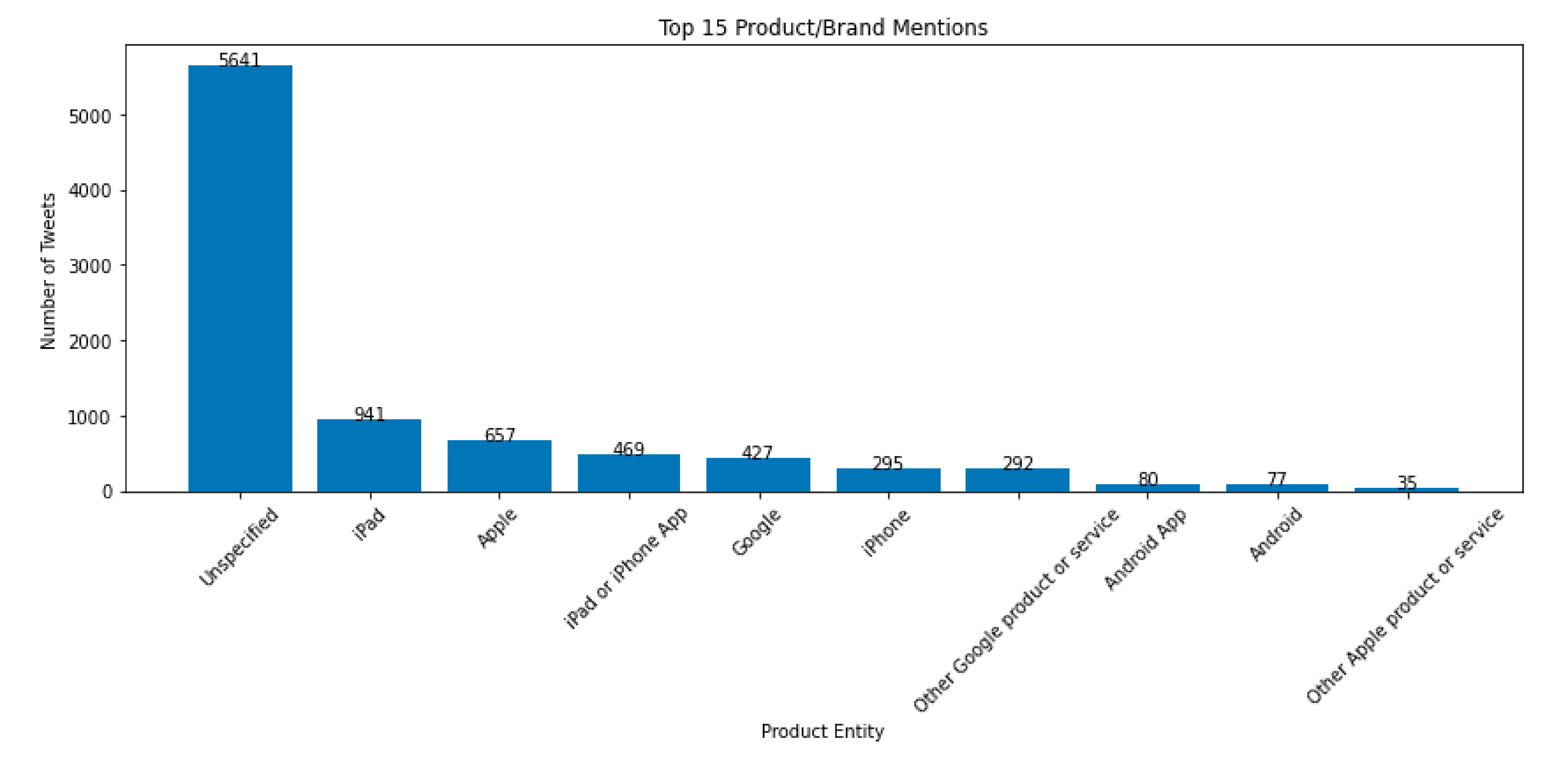
Clear Intentions of Sharing, Community, and Emotion.

Checking Outliers



The dataset exhibits relatively consistent vocabulary and thematic content, with no tweets being exceptionally distant from the overall corpus.

Top 15 Product/Brand Mention

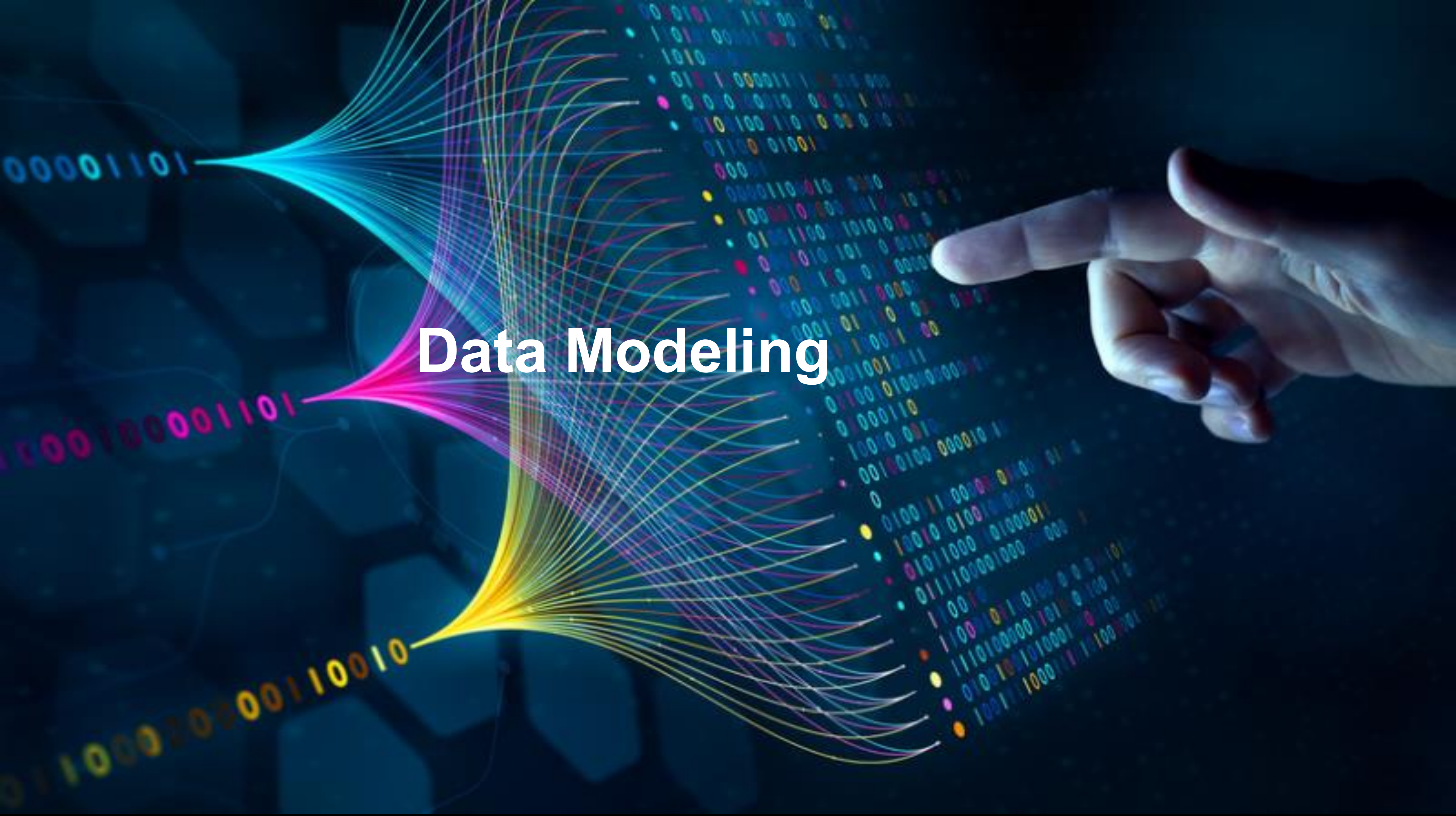


The most frequent words across all sentiment classes were product and brand names rather than emotional terms. Apple dominated Google

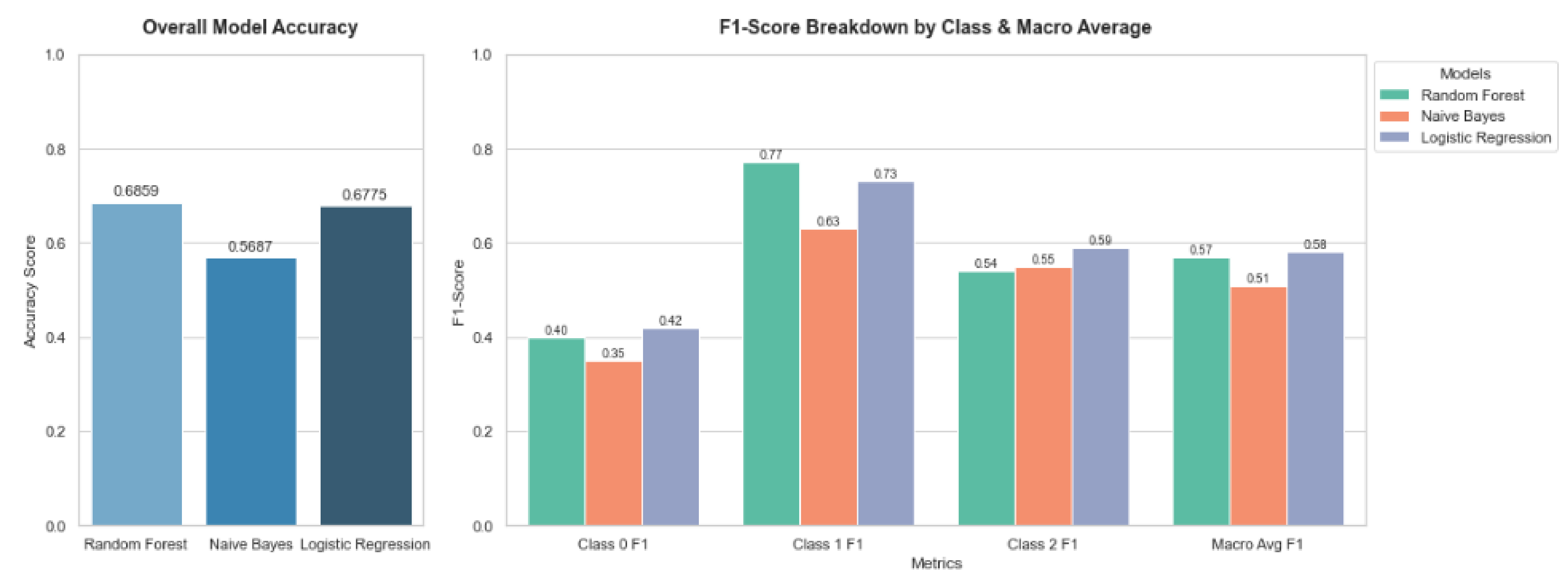
Checking Data Imbalance

SMOTE balanced dataset of 12,900 samples and increased the training set size by approximately 81%.
Thereby improving balanced decision and enhancing minority-class prediction performance.

Data Modeling



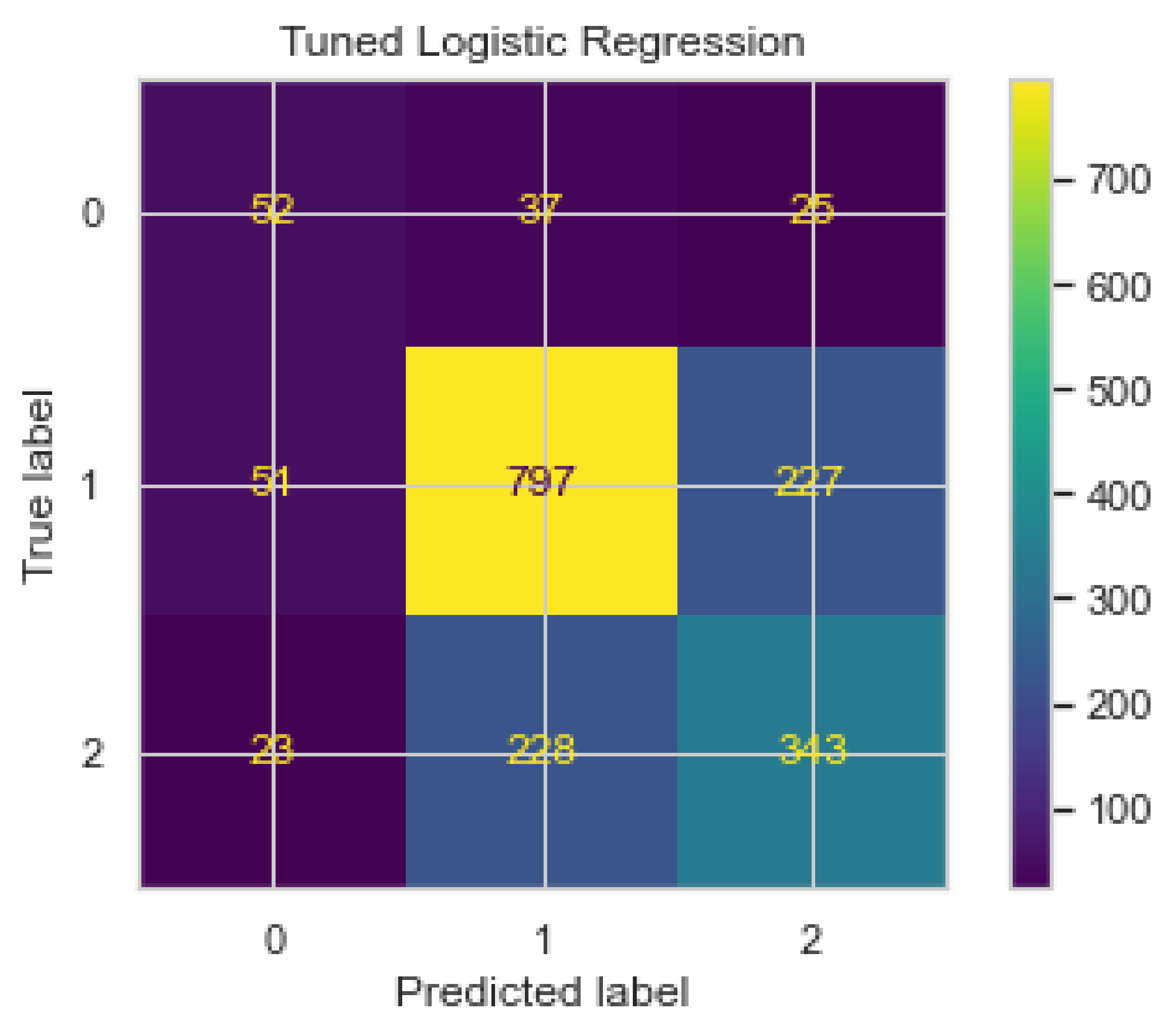
Model Comparison



The "Imbalanced Test Set" Trap: Even though SMOTE balanced instances, there still exists highly imbalance.

Logistic Regression was chosen for tuning because, it handles Class 0 significantly better and Higher Macro Average and lots of room for Optimization.

Tuned Logistic Regression.



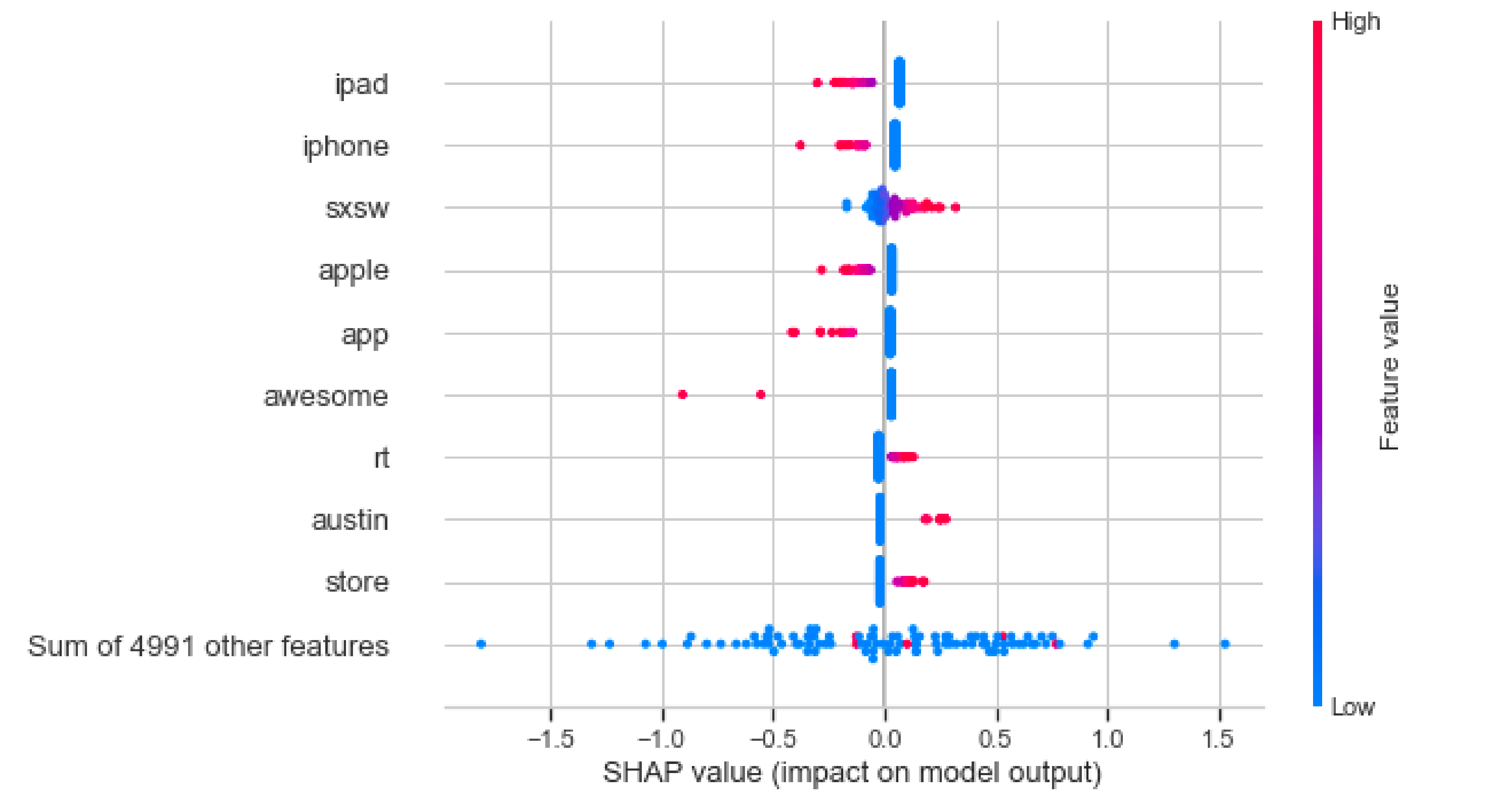
The Trade-Off (Accuracy vs. Balance):

Raw accuracy dropped 67.75% to 66.85%.

Macro Average F1-score increased from 0.58 to 0.59 confirming the model is better.

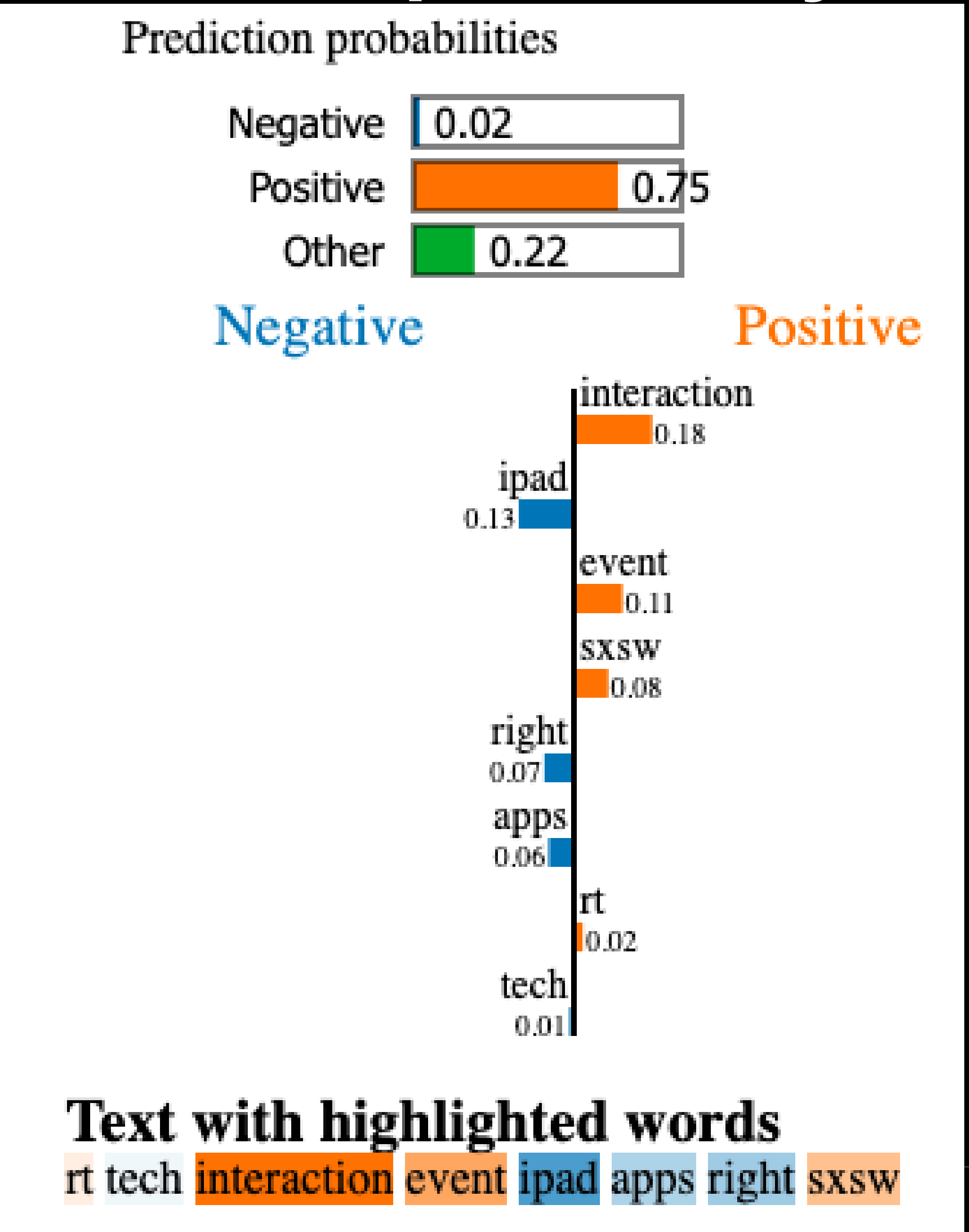
Class 1 (Majority) stabilized precision (0.75) and recall (0.74)). Class 2 precision and recall of 0.58.

SHAP Explainability



Words like "ipad", "iphone", "apple", and "app" have a strong negative impact, when they appear in a text.

Lime Explainablity.



The word "ipad" previously associated with negative emotions dragging predictions down.

In LIME when paired with positive content the model combines them to score it as a highly positive sentiment.

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Class 1 (Majority) stabilized precision (0.75) and recall (0.74)). Class 2 precision and recall of 0.58.

While SHAP shows how a word's meaning shifts depending on the company it keeps in a specific sentence

Insights

Business Perspective Insights

Deploy the Automated Alert Engine Immediately.

Prioritize the "Negative Class" Guardrails.

Build Live Corporate Brand Health Dashboards.

Exploit the Real-Time Competitive Edge.

Bridge the Communication Gap Between Tech and Business.

Optimize Human Resource Allocation to customer experience teams to follow up.

Questions?